Aarhus University
Cryptographic Computing
Group 2
Identity-Based Encryption from the Weil Pairing
(Proposal)

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1 Brief Summary

In regular public-key encryption schemes you need to know the public key of the people you want to talk to. This introduces a number of problems in terms of key management.

Identity Based Encryption (IBE) is the simplest case of functional encryption where we can encrypt messages that can only be decrypted by the owner of that email address. This requires the existence of some central authority that checks that we are the owner of some id and then provides us with a secret key corresponding to this identifier sk_{id}.

Now everyone can encrypt using some public parameter pp and our identity and compute $C = Enc(pp, id, m)$ in such a way that C reveals no information about m but the owner of sk_{id} can compute $m = Dec(sk_{id}, C)$.

2 Tasks

In this project our main tasks: read, understand, and implement a simple IBE scheme which was presented in [BF01] with use of *pairing*s as a black-box.